Supplemental Materials: Party or Climate? The Polarization of the US Public's Energy Preferences

Methods and Robustness Checks

Survey Sample

Despite the use of quotas, the sample from Cloud Research was not perfectly representative of the US population. Specifically, the sample was younger, whiter, more educated, and more aligned with the Democratic Party than the US population. To address this, I used the `survey` and `anesrake` packages in R to perform iterative raking to compute survey weights that align the sample with the US Census's 2023 American Community Survey (ACS) on key demographic variables, including age, race, gender, and education. Additionally, I used the Pew Research Center's National Public Opinion Reference Survey (NPORS) from 2025 to align the sample on political party affiliation.[1] The differences between the unweighted and weighted samples are shown in Table 1.

Table 1: Unweighted vs. Weighted Means on Demographic and Party Variables

	Unweighted	Weighted	Difference
Age (years)	43.624	47.287	3.662
Male	0.489	0.472	-0.016
White (non-Hispanic)	0.677	0.599	-0.077
College graduate	0.556	0.362	-0.194
Democrat	0.563	0.450	-0.113
Republican	0.287	0.460	0.173

[1] See here:https://www.pewresearch.org/politics/fact-sheet/party-affiliation-fact-sheet-npors/.

	Unweighted	Weighted	Difference

Note: Weights computed via iterative raking to ACS 2023 and Pew NPORS targets.

Robustness: Results Without Survey Weights

The analysis presented in the manuscript used the survey weights. To confirm that the main results are not a result of the weights, I re-estimate the preferred-mix models with unweighted OLS using the same specification (party * treatment for each energy source). Table 2 replicates the pairwise-differences table from the main text, and Table 3 replicates the difference-in-differences table. The estimates are substantively unchanged from the weighted analysis, indicating that the cue effects are likely not a product of the weights.

Table 2: Pairwise Differences in the Predicted Preferred Energy Mix (Unweighted)

Comparison	Fossil Fuels	Wind	Solar	Hydro	Nuclear
Gap (Dem - Rep): Control	-16.3***	10.2***	10.2***	2.2**	-6.2***
Gap (Dem - Rep): Climate	-16.7***	9.1***	10.3***	3.7***	-6.4***
Gap (Dem - Rep): Trump	-22.1***	11.0***	10.6***	3.4***	-2.9*
Republicans: Climate - Control	-1.1	1.4	-0.0	0.1	-0.3
Republicans: Trump - Control	7.0***	-1.4	-1.7	-1.0	-3.0*
Republicans: Trump - Climate	8.1***	-2.8**	-1.7	-1.1	-2.6†
Democrats: Climate - Control	-1.5	0.3	0.1	1.6**	-0.5
Democrats: Trump - Control	1.1	-0.5	-1.3	0.3	0.4
Democrats: Trump - Climate	2.7*	-0.8	-1.4	-1.4*	0.9

*Note: Replicates the pairwise-differences table from the main text using unweighted OLS models (no survey weights) in place of the weighted svyglm models. Cell entries are estimated differences in the predicted preferred share (percentage points). Partisan-gap rows are the Liberal Democrats – Conservative Republicans difference within each condition; within-party rows are the change for that party when moving from one condition to another. *** p < 0.001, ** p < 0.01, * p < 0.05, † p < 0.10.*

Table 3: Treatment Effects on the Preferred Energy Mix Across Political Beliefs (Unweighted)

Contrast	Fossil Fuels	Wind	Solar	Hydro	Nuclear
Omnibus F-test (any treatment effect on gap)	F = 5.99**	F = 1.17	F = 0.05	F = 1.33	F = 2.59†
Climate Cue vs Control	-0.37 [-4.03, 3.29]	-1.11 [-3.61, 1.38]	0.14 [-2.68, 2.96]	1.51 [-0.41, 3.44]	-0.17 [-3.53, 3.20]
Trump Cue vs Control	-5.85** [-9.54, -2.16]	0.86 [-1.66, 3.38]	0.43 [-2.42, 3.28]	1.21 [-0.73, 3.15]	3.35† [-0.05, 6.75]
Trump Cue vs Climate	-5.48** [-9.19, -1.77]	1.97 [-0.56, 4.50]	0.29 [-2.57, 3.15]	-0.30 [-2.25, 1.65]	3.52* [0.11, 6.92]

*Note: Replicates the difference-in-differences table from the main text using unweighted OLS models (no survey weights) in place of the weighted svyglm models. Top row shows the omnibus F-test for whether treatments jointly shift the partisan gap on each energy source. Subsequent rows show the estimated change in the partisan gap (Liberal Democrats – Conservative Republicans) when moving from one treatment condition to another, with 95% confidence intervals in brackets. *** p < 0.001, ** p < 0.01, * p < 0.05, † p < 0.10.*

Covariate Balance Across Conditions

Respondents were randomly assigned to the Trump, climate, or control conditions. Table 4 compares the demographic covariates across the three conditions to confirm that the random assignment produced comparable groups. Of the covariates examined, the conditions differed significantly only in terms of age and race; however, political beliefs, the variables of interest, were balanced across conditions.

Table 4: Covariate Balance Across Experimental Conditions

Variable	Control	Climate	Trump	p-value
Age (years)	42.63	44.51	43.76	0.019
Male	49.1%	49.2%	48.3%	0.915
White (non-Hispanic)	69.5%	63.5%	70.0%	0.002
Education (1–8)	4.07	4.03	4.03	0.793
Income (1–11)	4.71	4.61	4.55	0.415
College graduate	56.1%	54.7%	56.0%	0.772
Democrat	54.7%	55.8%	58.4%	0.208
Republican	29.5%	29.3%	27.4%	0.488

Variable	Control	Climate	Trump	p-value
Liberal Democrat	45.7%	47.6%	49.8%	0.171
Conservative Republican	25.3%	25.6%	22.5%	0.195

Note: Cell entries are unweighted condition means: percentages for binary indicators and means for continuous measures. The p-value tests for differences across the three conditions (chi-square for indicators, one-way ANOVA for continuous measures); non-significant values indicate the covariate is balanced across conditions, as expected under random assignment.

Robustness: Controlling for Demographic Covariates

Because of the differences in age and race across the conditions, I re-estimated the difference-in-differences models with age, sex, race, education, and income added as controls. Table 5 and Table 6 report largely the same results as the main-text pairwise-differences and difference-in-differences tables. The estimated cue effects on the partisan gap are substantively unchanged from what is shown in the manuscript, indicating that the results are likely not driven by the age and race imbalances across conditions.

Table 5: Pairwise Differences in the Predicted Preferred Energy Mix with Demographic Controls

Comparison	Fossil Fuels	Wind	Solar	Hydro	Nuclear
Gap (Dem - Rep): Control	-16.1***	10.3***	9.6***	1.8*	-5.6***
Gap (Dem - Rep): Climate	-18.6***	10.4***	10.5***	4.0***	-6.3***
Gap (Dem - Rep): Trump	-23.6***	11.9***	11.3***	3.4***	-3.2*
Republicans: Climate - Control	0.2	1.0	-1.0	-0.6	0.4
Republicans: Trump - Control	9.5***	-2.4*	-3.3*	-1.5	-2.3
Republicans: Trump - Climate	9.2***	-3.4**	-2.3†	-0.9	-2.7
Democrats: Climate - Control	-2.2*	1.0	-0.1	1.6*	-0.3
Democrats: Trump - Control	2.0†	-0.8	-1.6	0.1	0.2
Democrats: Trump - Climate	4.3***	-1.8†	-1.5	-1.5*	0.5

*Note: Replicates the pairwise-differences table from the main text, adding age, sex, race, education, and income as additive controls to each svyglm model. Cell entries are estimated differences in the predicted preferred share (percentage points). Partisan-gap rows are the Liberal Democrats – Conservative Republicans difference within each condition; within-party rows are the change for that party when moving from one condition to another (positive = the later condition is higher). Estimates use survey weights; standard errors and p-values are design-based. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$.*

Table 6: Treatment Effects on the Polarization of Preferred Energy Sources with Demographic Controls

Contrast	Fossil Fuels	Wind	Solar	Hydro	Nuclear
Omnibus F-test (any treatment effect on gap)	F = 4.03*	F = 0.78	F = 0.50	F = 1.81	F = 1.17
Climate Cue vs Control	-2.45 [-7.02, 2.13]	0.04 [-3.13, 3.21]	0.90 [-2.46, 4.26]	2.19† [-0.17, 4.55]	-0.69 [-5.08, 3.70]
Trump Cue vs Control	-7.43** [-12.56, -2.29]	1.60 [-1.30, 4.50]	1.72 [-1.65, 5.09]	1.64 [-0.68, 3.96]	2.47 [-1.66, 6.60]
Trump Cue vs Climate	-4.98† [-10.23, 0.27]	1.56 [-1.44, 4.55]	0.81 [-2.47, 4.10]	-0.55 [-2.89, 1.78]	3.16 [-1.22, 7.55]

*Note: Replicates the difference-in-differences table from the main text, adding age, sex, race, education, and income as additive controls to each svyglm model. The top row shows the omnibus F-test for whether treatments jointly shift the partisan gap on each energy source; subsequent rows show the estimated change in the partisan gap (Liberal Democrats – Conservative Republicans) when moving from one condition to another, with 95% confidence intervals in brackets. Estimates use survey weights; standard errors and confidence intervals are design-based. *** p < 0.001, ** p < 0.01, * p < 0.05, † p < 0.10.*

Preregistered Hypotheses

In addition to the hypotheses presented in the manuscript, I also preregistered other hypotheses that were not included in the analysis. First, were hypotheses regarding political beliefs on support/opposition to the climate change and Trump cues. Second, were hypotheses about the impacts of the cues on the preferred energy mix for all respondents (i.e., not split by political beliefs).

Cue Hypotheses

> Conservative Republicans will be more supportive of Trump's actions to increase the use of fossil fuels and decrease the use of renewable sources than those with other political beliefs
>
> Liberal Democrats will be more supportive of increasing the use of renewable sources to address climate change than those with other political beliefs

The results presented in Figure 1 in the manuscript show that these hypotheses were confirmed.

Energy Mix All Respondents Hypotheses

> Those who receive the Trump cue will support a higher percentage of fossil fuels than those in the climate change or control group

Those who receive the climate cue will support a higher percentage of renewable energy sources (wind, solar, hydropower) than those in the Trump or control group

Table 7 and Table 8 show the mean preferred percentage for each energy source under each treatment condition, pooled across all respondents, weighted (Table 7) and unweighted (Table 8). The p-value in each row tests whether the three condition means differ for that energy source (a design-based F-test for the weighted table, a one-way ANOVA for the unweighted table).

Table 7: Mean Preferred Energy Mix by Treatment Condition (Weighted)

Energy Source	Control	Climate	Trump	p-value
Fossil Fuels	22.9	22.3	27.0	0.001***
Wind	21.2	21.9	20.1	0.057†
Solar	24.0	23.3	22.7	0.234
Hydro	13.9	14.4	13.5	0.257
Nuclear	17.9	18.3	16.7	0.269

*Note: Cell entries are weighted mean preferred percentage allocations by treatment condition, pooled across all respondents (not split by political beliefs). The p-value tests whether the three condition means differ (design-based F-test). *** p < 0.001, ** p < 0.01, * p < 0.05, † p < 0.10.*

Table 8: Mean Preferred Energy Mix by Treatment Condition (Unweighted)

Energy Source	Control	Climate	Trump	p-value
Fossil Fuels	20.3	19.4	22.6	0.000***
Wind	22.3	23.1	22.0	0.112
Solar	25.1	25.0	24.7	0.745
Hydro	14.1	14.9	14.1	0.082†
Nuclear	18.1	17.5	16.6	0.099†

*Note: Cell entries are unweighted mean preferred percentage allocations by treatment condition, pooled across all respondents (not split by political beliefs). The p-value tests whether the three condition means differ (one-way ANOVA). *** p < 0.001, ** p < 0.01, * p < 0.05, † p < 0.10.*

As shown, respondents in the Trump condition preferred more fossil fuels than those in the other conditions, with or without survey weights. In addition, the climate condition showed some increases in renewable sources,

but these results were not robust, with p-values <0.10 and differences between the weighted and unweighted samples.